

Silicon Photonic Interferometer Design

Design proposal and simulation results

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I Introduction

I designed six TE devices to compare their free spectral range (FSR) and estimate the waveguide group index. The layout contains two Mach-Zehnder interferometers (MZIs), two Fabry-Pérot (FP) cavities, a Michelson interferometer, and a Michelson interferometer with an FP cavity in one arm, which I refer to as MFP. The different lengths allow me to compare propagation delay with the delay introduced by the Bragg gratings.

The MZI and Michelson spectra give a waveguide group index near 4.19. After including the Bragg response in the calculation, the FP cavities give 4.196 and 4.192, and the MFP gives 4.188. The larger values obtained from the bare-length formula are apparent indices. They include delay that the simple calculation assigns incorrectly to the straight waveguide.

Table I Devices in EBeam_weizhu.gds

Cell name	Length in the submitted layout
TE_MZI_30um	Arms 150.001 and 179.999 μm ; difference 29.998 μm
TE_MZI_150um	Arms 149.999 and 300.001 μm ; difference 150.002 μm
TE_FP_150um	Cavity waveguide 149.999 μm
TE_FP_250um	Cavity waveguide 249.999 μm
TE_MM_100um	Michelson arm difference 100 μm
TE_MFP_100um	Cavity waveguide 100 μm ; front grating length 3.17 μm

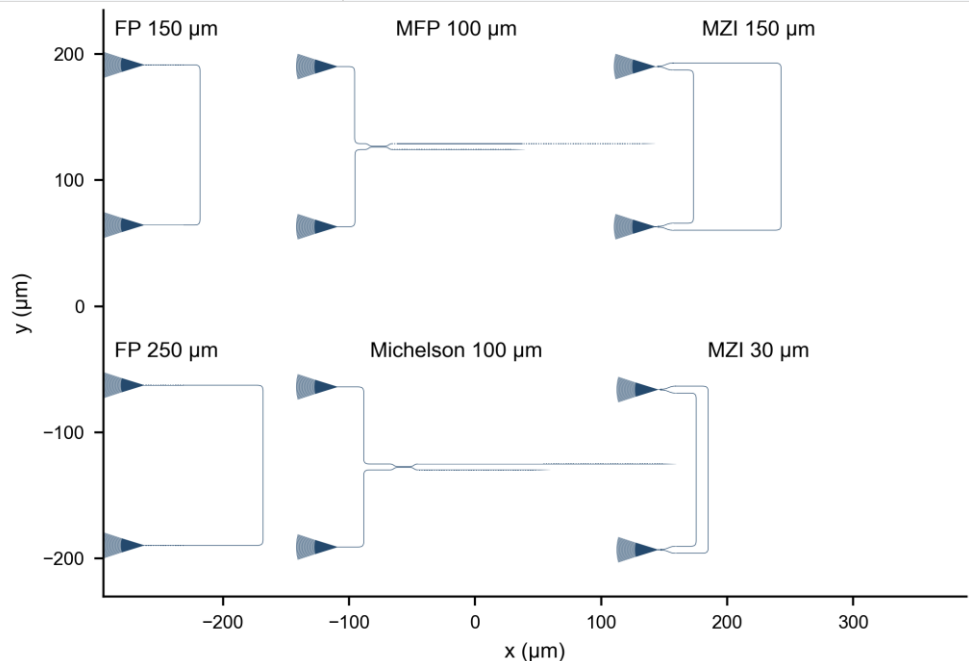


Figure 1 Silicon geometry from the submitted layout, top cell EBeam_weizhu_v1. Each device has one input and one output grating coupler. The vertical coupler spacing is 127 μm .

II Theory

The uniform strip waveguides have a nominal width of 500 nm and silicon thickness of 220 nm. At 1550 nm, the EBeam PDK TE model gives phase index $n_p = n_{\text{eff}} = 2.44553$ and group index $n_g = 4.19088$ [1]. These are different quantities: the phase index describes phase accumulation, while the group index describes propagation delay.

$$\beta = \frac{2\pi n_p}{\lambda}, \quad n_g = n_p - \lambda \frac{dn_p}{d\lambda} \quad (1)$$

For an ideal MZI with equal splitting and equal arm loss, the normalized transmission is

$$T_{\text{MZI}} = \frac{1 + \cos(\beta \Delta L)}{2} \quad (2)$$

Here ΔL is the arm-length difference. The frequency FSR and approximate wavelength FSR are

$$\Delta f_{\text{MZI}} = \frac{c}{n_g \Delta L}, \quad \Delta \lambda_{\text{MZI}} \approx \frac{\lambda^2}{n_g \Delta L} \quad (3)$$

Light travels to the mirrors and back in the Michelson. If the two mirrors contribute the same reflection phase, that phase cancels between the arms, giving

$$\Delta f_{\text{MM}} = \frac{c}{2n_g \Delta L}, \quad \Delta \lambda_{\text{MM}} \approx \frac{\lambda^2}{2n_g \Delta L} \quad (4)$$

For a Bragg FP cavity, the round-trip delay includes both the cavity waveguide and the two reflection delays τ_1 and τ_2 [5]. Locally in frequency,

$$\Delta f_{\text{FP}} \approx \frac{1}{\tau_{rt}}, \quad \tau_{rt} = \frac{2n_g L}{c} + \tau_1 + \tau_2 \quad (5)$$

L is the waveguide length between the gratings, and c is the speed of light in vacuum. Applying the bare-waveguide formula to this FSR gives an apparent index:

$$n_{g,app} = \frac{c}{2L\Delta f_{\text{FP}}} \approx n_g + \frac{c(\tau_1 + \tau_2)}{2L} \quad (6)$$

An apparent index above 4.19 can therefore result from mirror delay without a change in the strip-waveguide index. Ignoring that delay gives FP estimates of 1.91 nm for 150 μm and 1.15 nm for 250 μm at 1550 nm. Equations 3 and 4 give 19.11 and 3.82 nm for the MZIs, and 2.87 nm for the Michelson.

The MFP combines a cavity-reflected field with a reference-arm field. Its fringes depend on their relative phase, including the frequency-dependent cavity reflection phase [6]. I use the complete complex cavity reflection in its corrected calculation. FSR alone does not determine the absolute phase index n_p .

III Modelling and simulation

I simulated the circuits in Lumerical INTERCONNECT and saved one text export per device on 5 October 2026. The six files contain 5,000 wavelength and Y pairs over 1499.02-1601.00 nm, with about 0.02 nm wavelength spacing near 1550 nm. Figure 2 shows the saved gain traces without smoothing or normalization. The Optical Network Analyzer provides the circuit response [3].

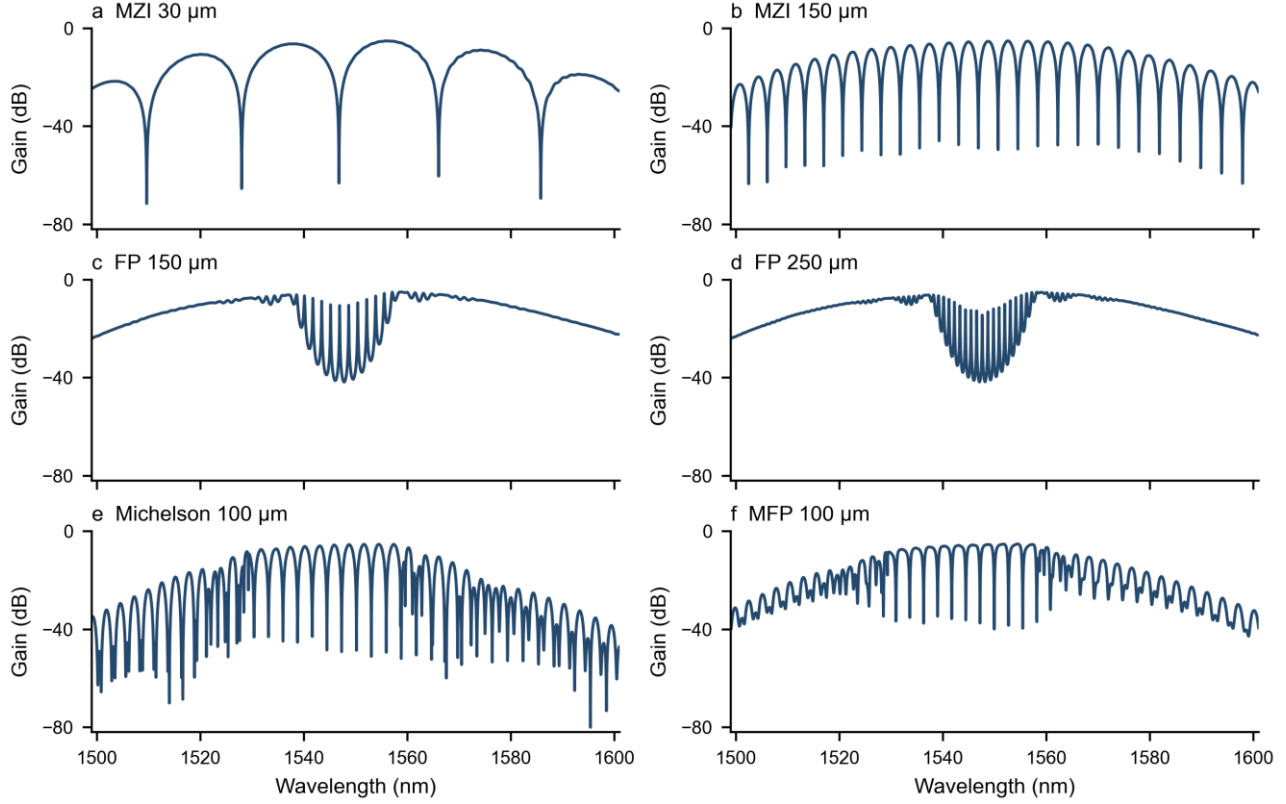


Figure 2 Full saved INTERCONNECT spectra. Panels a and b show the MZIs, c and d the FP cavities, e the Michelson, and f the MFP. Gain includes the grating couplers and other circuit components.

The GDS metadata lists a Bragg period of 317 nm for every grating. The standalone FP cavities have 100-period mirrors with 50 nm corrugation. The Michelson has two 300-period mirrors with 200 nm corrugation. The MFP has a 10-period front mirror, a 300-period rear mirror, and a 300-period reference mirror, all with 200 nm corrugation. Its front grating is $10 \times 317 \text{ nm} = 3.17 \text{ } \mu\text{m}$ long.

The MZIs use Y branches as the splitter and combiner. The Michelson and MFP use an EBeam directional coupler with 500 nm waveguide width, 200 nm gap, 10 μm coupling length, and 5 μm radius. The routed waveguides also have a 5 μm bend radius. These dimensions come from the submitted GDS; the saved gain files do not record the complete INTERCONNECT circuit settings or complex reflection coefficients. Bragg-grating models include a wavelength-dependent reflection response [4].

The earlier waveguide export wg.mat contains ten samples of effective index and group velocity. Linear interpolation gives phase index 2.4462 and group index 4.2036 at 1550 nm. This is a separate waveguide result. I use the PDK group index of 4.19088 for the circuit FSR estimates.

Data sources MZI layout test folder: TE_MZI_30um.txt, TE_MZI_150um.txt, TE_FP_150.txt, TE_FP_250.txt, TE_MM_100um.txt, and TE_MFP_100um.txt.

Simulated free spectral range

I used consecutive minima for the MZIs and reflecting interferometers, and transmission maxima for the FP cavities. Figure 3 marks the selected features. Most are in the 1542-1555 nm window; the 30 μm MZI uses 1520-1575 nm because its fringes are farther apart.

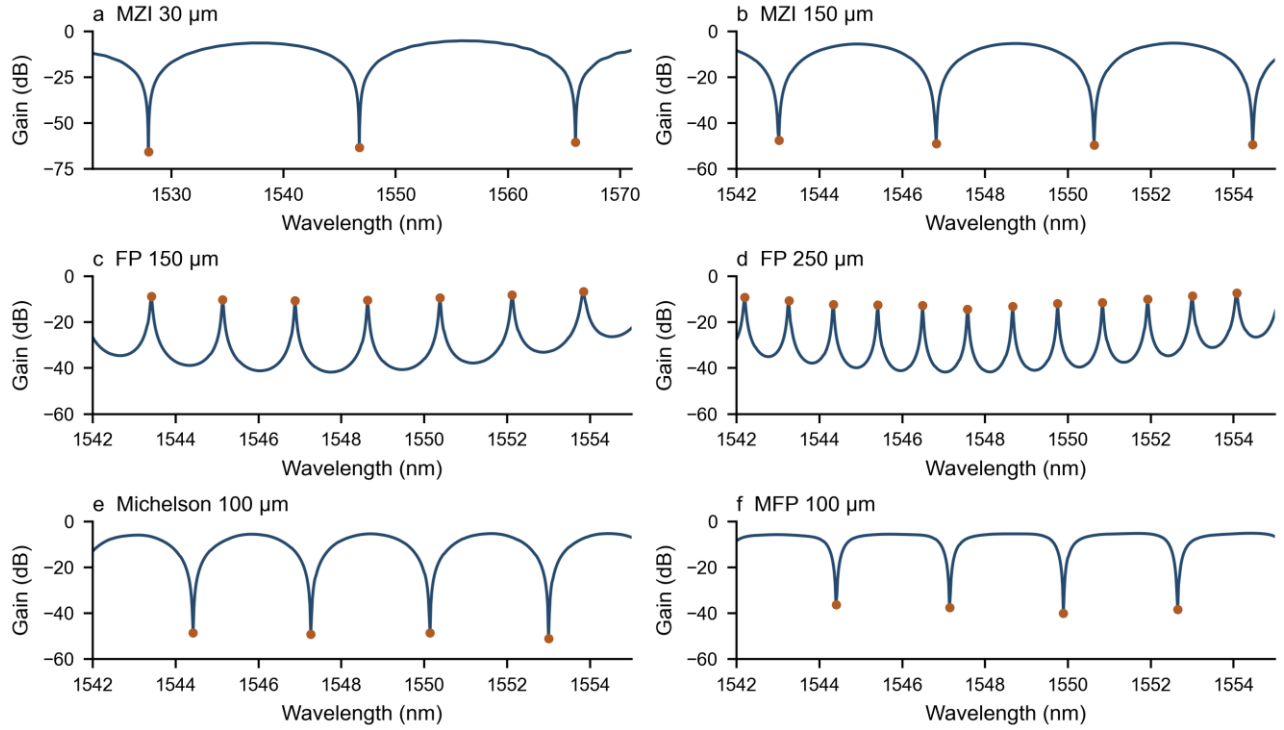


Figure 3 Spectral windows used for the FSR calculation. Orange markers identify the selected features. Panel a uses a wider wavelength range than the other panels.

Table II Original FSR calculation and model-corrected group index

Device	Saved FSR	Δf in GHz	Bare-length n_g^1	Corrected n_g^2
MZI 30 μm	19.03 nm	2384.3	4.192	4.192
MZI 150 μm	3.81 nm	476.6	4.193	4.193
Michelson 100 μm	2.86 nm	357.5	4.193	4.193
FP 150 μm	1.74 nm	217.3	4.599	4.196
FP 250 μm	1.08 nm	135.1	4.438	4.192
MFP 100 μm	2.75 nm	343.8	4.360	4.188

¹ MZI values use $c/(\Delta L \Delta f)$; the other rows use $c/(2L \Delta f)$. Individual FP and MFP bare-length values are apparent indices. ² FP and MFP values use the nominal PDK Bragg response and a fit to the saved power spectrum. MZI and Michelson values need no cavity correction under the stated assumptions.

I converted each selected wavelength to frequency using $f = c/\lambda$ and averaged the adjacent frequency intervals for the original FSR calculation. The MZI and Michelson indices agree with the PDK value to about 0.06%. The corrected FP and MFP values agree to within about 0.12%.

The MFP minima at 1544.40, 1547.14, 1549.89, and 1552.65 nm give 2.75 nm mean spacing. For the corrected values, I fitted all 638 saved samples in the 1542-1555 nm window with the group index as a free parameter. The grating coupling coefficient, Bragg wavelength, and grating loss were fixed from the installed component model. A smooth envelope accounts for the couplers and overall gain. Appendix A compares the fits with the saved data.

IV Fabrication

The design uses the EBeam PDK for the 220 nm silicon platform. I placed 12 TE grating couplers in six input-output pairs at 127 μm pitch. Each MZI has one input and one output, with Y branches connecting its two internal arms. Four EBeam terminators are placed at the unused far ends of the Michelson and MFP gratings.

The GDS contains 78 optical pins at 39 matching locations. It matches the submitted EBeam_weizhu.gds, merged in pull request 370 on 4 October 2026 [2]. The automated checks for that layout submission passed. Fabricated-device measurements are not available yet.

V Experimental data

After fabrication, I plan to sweep a TE-polarized laser near 1550 nm and record the output power. I will compare the MZI and Michelson group indices, then examine the FP pair using the delay difference. The MFP measurement will require its grating and cavity phase model as well as fringe spacing. All results reported here are from simulation exports.

VI Analysis

The MZI and Michelson provide the most direct waveguide-index estimates in this layout. Their values cluster around 4.193, close to the PDK reference. The two identical Michelson mirrors approximately cancel their common reflection phase, while the FP mirrors add their delays within each cavity.

The individual FP apparent indices, 4.599 and 4.438, assign the entire round-trip delay to the straight cavity. The missing contribution is the sum of the two mirror reflection delays. Using the frequency-dependent reflection phase of the nominal Bragg model in Equation 5 removes that contribution. Fitting the saved central power traces gives $n_g = 4.196$ for the 150 μm cavity and 4.192 for the 250 μm cavity.

As a separate cross-check, the cavities use identical mirrors and differ in length by 100 μm . Their FSRs give round-trip delays of 4.602 and 7.402 ps. Assuming approximately equal mirror delay over the selected windows, subtraction gives

$$n_{g,pair} \approx \frac{c}{2(L_{250} - L_{150})} \left[\frac{1}{\Delta f_{250}} - \frac{1}{\Delta f_{150}} \right] = 4.198 \quad (7)$$

This is within about 0.16% of the PDK reference. The common-delay approximation is not exact because the two cavities sample slightly different frequencies. The individual model fits include the changing mirror phase rather than assuming a constant penetration length.

Using $n_g = 4.19088$, the uncorrected FSRs correspond to effective cavity lengths of 164.6 and 264.8 μm , about 15 μm above each straight-waveguide length in total. This is consistent with extra reflection delay [5]. It does not mean that the straight waveguide has $n_g = 4.599$ or 4.438, or that each mirror adds 15 μm .

MFP correction

The MFP needs the reflection of the complete cavity, not just the phase of a 100 μm waveguide. In the GDS, both the reference grating and the cavity front grating begin at the coupler outputs. The rear mirror is 103.17 μm along the cavity arm: 3.17 μm of front grating plus the 100 μm straight cavity.

Let r_f and r'_f be reflection from the left and right of the front grating, t_f its reciprocal transmission, and r_b reflection from the rear grating. With $u = \exp(i\beta L)$, the cavity reflection is

$$r_{FP} = r_f + \frac{t_f^2 r_b u^2}{1 - r'_f r_b u^2} \quad (8)$$

The two reflected arms combine at the output coupler. For the opposite input-output ports used here, the modelled power is proportional to $|r_{FP} + r_{ref}|^2$, with a smooth gain envelope. The front grating contributes both direct reflection and transmission into the cavity; the denominator in Equation 8 includes repeated cavity round trips.

The bare 100 μm formula gives $n_{g,app} = 4.360$. Adding the physical front-grating length alone reduces the apparent value to 4.226 under a uniform-index approximation. That still omits the distributed grating response and repeated reflections. Including those effects in the complex-field model gives the corrected estimate $n_g = 4.188$.

This value is about 0.06% below the PDK reference. The fit uses the same free group-index parameter for propagation and grating detuning, as assumed by the component model; it does not fix the result to 4.19088. The central spectrum is consistent with the intended MFP topology within this model. The complicated response outside the fitted window is not assessed by this correction.

Phase index and group index

For the uniform strip segments, the circuit model uses phase index $n_p = 2.44553$ and group index $n_g = 4.19088$ at 1550 nm. The separate waveguide export gives 2.4462 and 4.2036. These phase-index values come from waveguide models; the fringe-spacing analysis measures delay and cannot independently determine absolute n_p .

VII Conclusion

The MZI, Michelson, and corrected FP and MFP results all give n_g close to 4.19. The apparent disagreement came from omitting the Bragg reflection and cavity phase, with the MFP also requiring its front-grating transmission. The phase-corrected estimates depend on the nominal component model and the saved spectral sampling. They support the central response of the design; they do not constitute fabricated-device measurements.

References

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- [3] Ansys Optics, Optical Network Analyzer ONA INTERCONNECT Element. <https://optics.ansys.com/hc/en-us/articles/360036617973>
- [4] Ansys Optics, Bragg Grating WBG INTERCONNECT Element. <https://optics.ansys.com/hc/en-us/articles/360036108114>
- [5] Y. O. Barmenkov et al., Effective length of short Fabry-Perot cavity formed by uniform fiber Bragg gratings, Optics Express 14, 6394-6399 (2006). <https://doi.org/10.1364/OE.14.006394>
- [6] J.-C. Diels and W. Rudolph, Ultrashort Laser Pulse Phenomena, Section 2.3.2. https://dielslab.unm.edu/sites/default/files/book_ultrafast_ch1_and_2.pdf

Appendix A Phase model and fit validation

I reconstructed the uniform Bragg response from the installed ebeam_bragg_te1550.ice setup script [1] using its rectangular-grating calibration. The geometric period is 317 nm. The group index is fitted freely in both waveguide propagation and grating detuning; the remaining grating parameters are fixed.

Corrugation	Bragg wavelength	Signed κ	Grating length
50 nm	1547.7045 nm	$+75,375.25 \text{ m}^{-1}$	31.70 μm for 100 periods
200 nm	1543.8124 nm	$-159,056 \text{ m}^{-1}$	3.17 / 95.10 μm for 10 / 300 periods

The grating loss is 590 dB/m. I used the standard uniform coupled-mode reflection and transmission with complex detuning to include this loss. The FP model combines two mirrors and the cavity propagation. The MFP uses Equation 8 and interferes that reflection with the 300-period reference mirror.

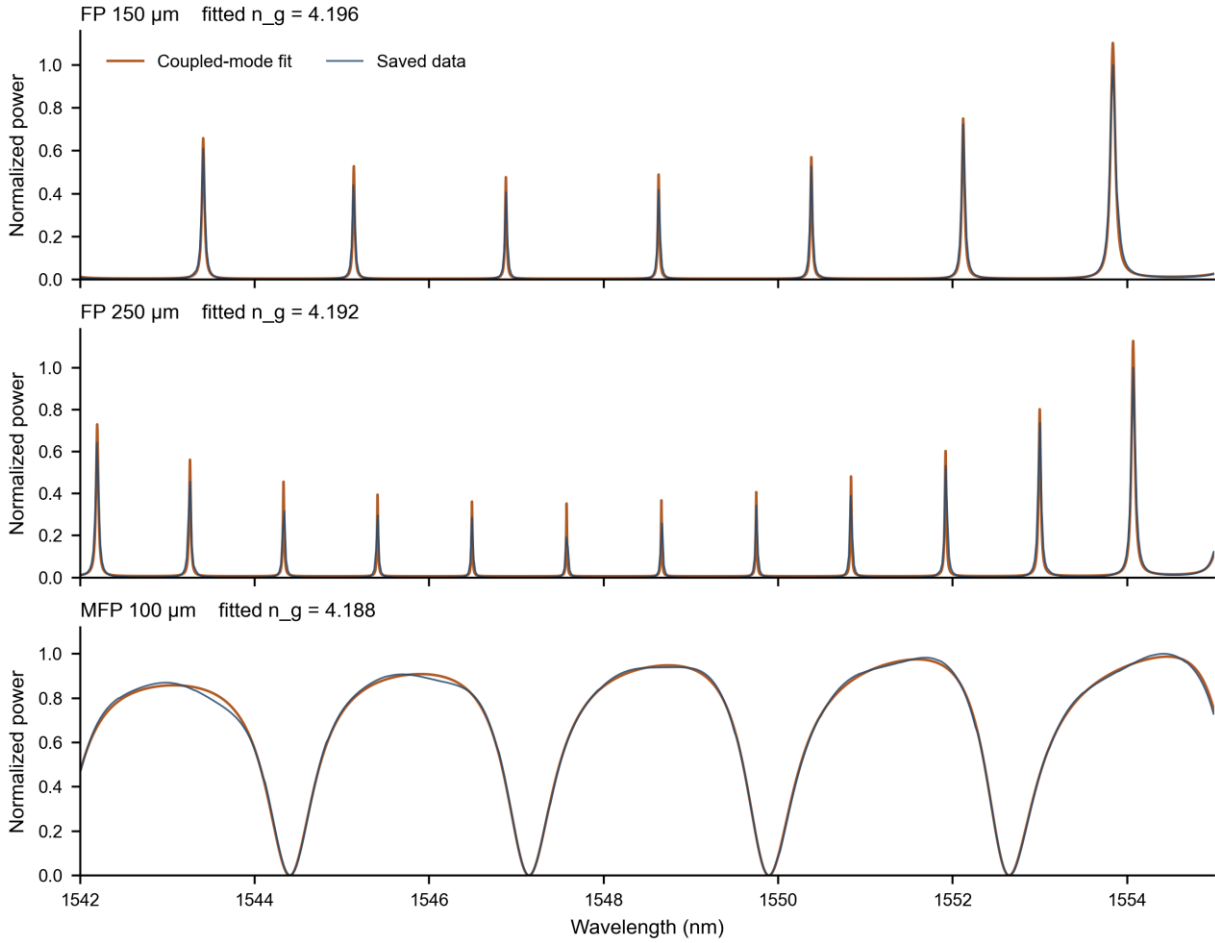


Figure 4 Phase-model fits compared with the saved central spectra. Power is normalized to the largest saved value in each window for fitting and display only; Figures 2 and 3 retain the original gain values. Narrow FP maxima between saved samples can exceed the sampled maximum.

The RMS differences are 1.29%, 1.93%, and 1.05% of normalized full-scale power for FP 150 μm , FP 250 μm , and MFP, respectively. Fits started from different group-index guesses converge to the same values. Fits to locally refined feature centres provide a cross-check within about 0.004 in n_g .

A phase offset, smooth multiplicative envelope, and small additive background are fitted with n_g . The quoted three decimals aid comparison rather than establish that level of accuracy. Grating calibration, the assumed common group index, and finite wavelength sampling limit the extraction. These corrections use the saved gain data and nominal model; no new INTERCONNECT run or measured complex phase is claimed.